A2 · Unit 5 · WBI15 | Respiration, Internal Environment, Coordination and Gene Technology

Control of the Internal Environment

Body cells work only if the conditions around them stay within narrow limits. This page covers how heart rate, cardiac output and breathing respond to exercise, then feedback and homeostasis, thermoregulation, the kidney, ADH, and how hormones switch genes on and off.

SPEC 7.13 Cardiac output; control of heart rate and ventilation

WHAT THIS SPEC POINT ASKS

(i) Calculate cardiac output. (ii) Understand how changes in ventilation and cardiac output deliver O_2 to the tissues and remove CO_2 quickly, including how heart rate and ventilation rate are controlled by the cardiovascular control centre and the ventilation centre in the medulla oblongata.

Cardiac output

Cardiac output = stroke volume × heart rate

- **Stroke volume**: volume of blood pumped out of the left ventricle in one beat (cm^3).
- **Heart rate**: beats per minute.
- **Cardiac output**: volume of blood pumped per minute ($cm^3\ min^{-1}$ or $dm^3\ min^{-1}$). About 5 $dm^3\ min^{-1}$ at rest; 20–30 $dm^3\ min^{-1}$ in hard exercise.

WORKED EXAMPLE

Worked example: cardiac output

One cardiac cycle takes 0.80 s, so heart rate = 60 ÷ 0.80 = 75 beats min^{-1}. Stroke volume = 72 cm^3.

Cardiac output = 72 × 75 = 5400 $cm^3\ min^{-1}$ = **5.4 $dm^3\ min^{-1}$** (÷ 1000).

Rearranged: cardiac output 4500 $cm^3\ min^{-1}$ and heart rate 51 beats min^{-1} give stroke volume = 4500 ÷ 51 = **88.24 cm^3** (WBI15 Jun 2021).

Percentage increase from rest (4000 $cm^3\ min^{-1}$) to exercise (21 250 $cm^3\ min^{-1}$) = (21 250 − 4000) ÷ 4000 × 100 = **431%**.

Change in cardiac output after a swim

1. At rest: heart rate 65 bpm and stroke volume 70 cm3, so cardiac output = 65 × 70 = 4550 cm3 min-1.
2. After the swim: cardiac output is 10 800 cm3 min-1 and heart rate is 120 bpm.
3. Change in cardiac output = 10 800 − 4550 = 6250 cm3 min-1.
4. Stroke volume after the swim = 10 800 ÷ 120 = 90 cm3.

Rearranging the same equation gives the missing variable. Both rising heart rate and rising stroke volume account for the increase.

- **Trained athletes** have a larger, stronger left ventricle, so a bigger stroke volume. They get the same resting cardiac output with a **lower resting heart rate**, and reach a higher maximum cardiac output.

The first interactive diagram at the foot of the page shows heart rate, stroke volume and cardiac output together, for an untrained person and a trained athlete.

Why cardiac output and ventilation rise together in exercise

- Working muscles respire faster, so they use more **O_2** and make more **CO_2** (and heat).
- Higher **cardiac output** delivers O_2 and glucose faster and carries CO_2 away faster.
- Higher **ventilation** (faster, deeper breathing) keeps alveolar air high in O_2 and low in CO_2, so the diffusion gradients across the alveolar walls stay steep.
- Blood is redistributed: arterioles supplying skeletal muscle (and the skin, for heat loss) dilate, while those supplying the gut and kidneys constrict.

Control of heart rate

- The **cardiovascular control centre** is in the **medulla oblongata**.
- It receives impulses from:
  - **chemoreceptors** in the aortic bodies, carotid bodies and medulla, which detect **rising CO_2 / falling pH** (CO_2 dissolves in plasma, forming carbonic acid, which releases H^+);
  - **baroreceptors** in the aortic arch and carotid sinuses, which detect **blood pressure**;
  - stretch receptors in muscles and joints, which detect movement at the start of exercise.
- It responds by changing the impulses sent to the **SAN**:
  - more impulses along the **sympathetic (accelerator) nerve** → noradrenaline released at the SAN → SAN **depolarises more frequently** → heart rate **rises** (and the ventricles contract more strongly, raising stroke volume);
  - more impulses along the **parasympathetic vagus nerve** → acetylcholine released at the SAN → SAN fires less often → heart rate **falls**.

Change detected	Receptors	More impulses along	Heart rate
CO_2 rises, pH falls (exercise)	chemoreceptors	sympathetic nerve to SAN	rises
CO_2 falls, pH rises (after exercise)	chemoreceptors	vagus (parasympathetic) nerve to SAN	falls
Blood pressure rises	baroreceptors	vagus nerve to SAN	falls
Blood pressure falls	baroreceptors	sympathetic nerve to SAN	rises

Control of ventilation

- The **ventilation centre** is in the **medulla oblongata**. It sends impulses rhythmically to the **external intercostal muscles** and **diaphragm**, which contract to breathe in.
- When CO_2 rises (pH falls), **chemoreceptors** in the medulla and in the carotid and aortic bodies send more impulses to the ventilation centre. It sends **more frequent impulses** to the intercostal muscles and diaphragm, so breathing becomes **faster and deeper**.
- **Stretch receptors** in the walls of the bronchioles send impulses when the lungs inflate. These **inhibit** inspiration, so breathing out follows and the lungs don't over-inflate.
- At rest, breathing out is mostly passive: the muscles relax and the lungs recoil elastically.
- When CO_2 falls back after exercise, fewer impulses reach the breathing muscles, and the rate and depth fall.

MARK-SCHEME WORDING

- Heart rate rising (any 4): "**increase in blood CO_2 / decrease in pH**"; "detected by **chemoreceptors**"; "impulses to the **cardiovascular control centre** in the **medulla oblongata**"; "**more impulses** via the **sympathetic** nerve to the **SAN**"; "SAN depolarises **more frequently**"; "heart muscle contracts more often". Baroreceptors and adrenaline are accepted alternatives.
- Slowing: pH rises after exercise; impulses along the **parasympathetic / vagus** nerve slow the SAN.
- Ventilation: "**chemoreceptors**"; "impulses to the **ventilation centre** in the medulla"; "**more impulses to the intercostal muscles and diaphragm**"; "increased rate (and depth) of breathing".
- Cardiac output: "heart rate × stroke volume"; it "measures the volume pumped per minute, so allows for different heart sizes". Athletes: "higher stroke volume, so lower resting heart rate for the same output".
- Write "impulses", never "signals" or "messages". Write "more" impulses.

Often asked as: cardiac output, stroke volume or heart rate calculations (with units and s.f.); how heart rate rises in exercise or falls afterwards; how breathing rate is controlled; 6-mark questions on the heart and exercise (athletes v non-athletes); blood redistribution to skin and muscles.

COMMON MISTAKE

- Describing what happens **during** exercise when the question asks about **recovery** (or surfacing after a dive).
- Saying the vagus nerve "sends impulses to the medulla". It carries impulses **from the medulla to the SAN**.
- Saying impulses reach the diaphragm along "sympathetic nerves". The breathing muscles are stimulated by motor neurones (intercostal and phrenic nerves), not the autonomic system.
- Finishing a ventilation answer by talking about heart rate.
- Thinking breathing rate rises *because* heart rate rises (or the reverse). Both respond to the same rise in CO_2.
- Forgetting stroke volume: cardiac output rises in exercise because **both** heart rate and stroke volume rise.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Cardiac index** = cardiac output ÷ body surface area, in $dm^3\ min^{-1}\ m^{-2}$ (WBI15 Jun 2022).
- **Alveolar ventilation** = (tidal volume − dead space) × breathing rate: (500 − 150) × 12 = 4200 $cm^3\ min^{-1}$ = 4.2 $dm^3\ min^{-1}$; × 60 = **250 $dm^3\ h^{-1}$** to 2 s.f. Units missing or wrong capped it at 2 marks (WBI15 Jan 2022).
- Lung stretch receptors inhibiting inspiration (WBI15 Jun 2022 article).
- Diving seals: the heart rate slows a lot during a dive (WBI15 Oct 2025).

SPEC 7.14 Adrenaline and the fight-or-flight response

WHAT THIS SPEC POINT ASKS

Understand the role of adrenaline in the fight-or-flight response.

What triggers it

- A threat (or the anticipation of exercise) is detected. The hypothalamus activates the **sympathetic nervous system**.
- Sympathetic neurones stimulate the **adrenal medulla** (the inner part of each adrenal gland, on top of the kidneys) to secrete **adrenaline** into the **blood**.
- Adrenaline reaches every tissue, but it only affects **target cells** that have **adrenaline receptors** on their cell surface membranes.

How it acts on a target cell

- Adrenaline can't cross the membrane, so it **binds to receptors on the cell surface membrane**.
- This activates an enzyme (adenylyl cyclase) that makes a **second messenger**, **cAMP**, inside the cell. cAMP activates a cascade of enzymes, so one adrenaline molecule has a large, fast effect. Peptide hormones work the same way (7.22).
- Adrenaline is an **amine** hormone (made from the amino acid tyrosine), not a peptide or a steroid.

The effects: ready to fight or run

Effect	How it helps
Heart rate and **stroke volume** rise (adrenaline acts on the SAN and the ventricle muscle)	higher cardiac output: O_2 and glucose reach the muscles faster
Vasodilation of arterioles to skeletal muscle; **vasoconstriction** of arterioles to the gut and skin	blood redirected to the muscles
Bronchioles dilate; breathing becomes faster and deeper	more O_2 in, more CO_2 out
Liver **glycogen → glucose**	more blood glucose for respiration
Pupils dilate (radial muscles of the iris contract, circular muscles relax)	more light enters the eye
Digestion slows (gut muscle relaxes)	blood and energy diverted to the muscles
Greater mental alertness	faster reactions

MARK-SCHEME WORDING

- Role (any 3): "released **into the blood** from the **adrenal glands**"; "**binds to receptors** on target cells / organs"; then named effects: "increased **heart rate**", "**vasodilation** in muscle", "increased **ventilation** rate", "**pupil dilation**", "relaxation of gut muscle / decreased digestion", "**glycogen to glucose** in the liver".
- Pupil: "sympathetic / motor neurones stimulated", "**circular** muscles **relax**", "**radial** muscles **contract**".
- Combination data (heart rate of *Daphnia*): heart rate rose with adrenaline and fell with nicotine; highest rate with the most adrenaline and least nicotine. Use the **combination** of chemicals, as the question asked (WBI15 Oct 2023).

Often asked as: MCQs naming the hormone and its effects on breathing and heart rate; describe the role of adrenaline in fight or flight; adrenaline and the pupil; 6-mark question on ageing and the heart's response to adrenaline.

COMMON MISTAKE

- Leaving out the two points few candidates gave: adrenaline is **released into the blood from the adrenal glands**, and it **binds to receptors**.
- Calling adrenaline a **peptide** hormone, or saying it enters the cell. It's an amine that acts at the cell surface.
- Telling the whole iris reflex story when the question is about adrenaline.
- Confusing adrenaline (a hormone from the adrenal medulla) with noradrenaline (the neurotransmitter released by sympathetic neurones at the SAN).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Adrenaline acting directly on the **SAN** is credited as an alternative route for raising heart rate (WBI15 Oct 2022, Oct 2023).
- Ageing reduces the heart's response to adrenaline: use all three graphs (heart rate, cardiac index, blood pressure), compare older and younger people, and comment on the error bars (WBI15 Jun 2022, 6-mark).
- Adrenaline also counted as a neurotransmitter in an MCQ pair (WBI15 Jan 2026).

SPEC 7.15 Core Practical 17: exercise and spirometer traces

WHAT THIS SPEC POINT ASKS

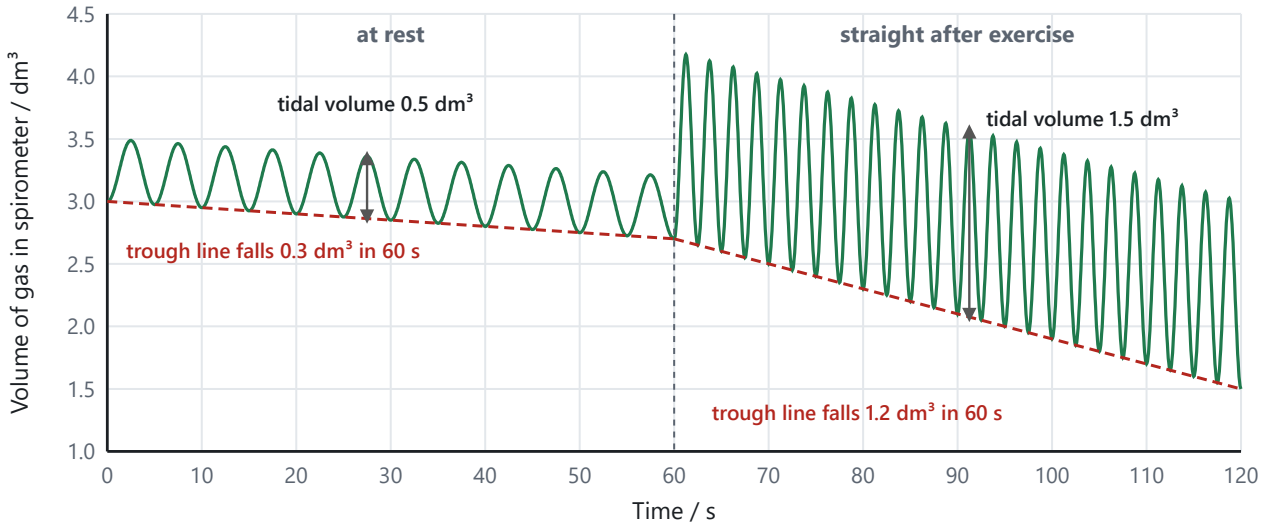
Investigate the effects of exercise on tidal volume, breathing rate, minute ventilation and oxygen consumption, using data from spirometer traces.

How a spirometer works

- A chamber of air (or medical-grade oxygen) floats over water. The subject breathes in and out of it through a mouthpiece, wearing a **nose clip**. The chamber rises and falls, and a pen or sensor records a trace of volume against time.
- **Soda lime** absorbs the CO_2 breathed out, so it isn't breathed back in. As the subject uses O_2, the **total volume** of gas in the chamber **falls**, so the trace slopes downwards. The **slope** gives the rate of O_2 consumption.

Reading the trace

Measurement	Meaning	How to read it
Tidal volume	volume of air breathed in (or out) in one normal breath	height from a **peak to the next trough**
Breathing rate	breaths per minute	count the peaks in a known time, then convert to per minute
Minute ventilation	volume of air breathed in per minute	tidal volume × breathing rate
O_2 consumption	volume of O_2 used per minute	fall in a line through the troughs (or peaks) ÷ time
Vital capacity	largest volume breathed out after the deepest possible breath in	biggest peak-to-trough height



A spirometer trace. After exercise the breaths are deeper (bigger tidal volume) and more frequent, and the trace falls more steeply because more O_2 is being used.

WORKED EXAMPLE

Worked example: reading the trace above

At rest: 12 peaks in 60 s → breathing rate = **12 breaths min^{-1}**. Peak to trough = **0.5 dm³** (tidal volume).

Minute ventilation = 0.5 × 12 = **6.0 dm³ min^{-1}**.

The trough line falls from 3.0 to 2.7 dm³ in 60 s → O_2 consumption = **0.3 dm³ min^{-1}** (5.0 × 10^{-3} dm³ s^{-1}).

After exercise: 24 peaks in 60 s → **24 breaths min^{-1}**; tidal volume = **1.5 dm^3**; minute ventilation = 1.5 × 24 = **36 dm^3 min^{-1}**, six times the resting value.

The trough line falls from 2.7 to 1.5 dm^3 in 60 s → O_2 consumption = **1.2 dm^3 min^{-1}**, four times the resting rate.

CORE PRACTICAL

Core practical 17: rest and exercise

1. Check the subject is healthy (no asthma, heart or breathing problems) and has given **informed consent**.
2. Fill the spirometer. Check the **soda lime** is fresh. Fit a **sterilised or new mouthpiece** and a **nose clip**.
3. Let the subject breathe normally to get used to the apparatus, then record for **1–2 minutes at rest**.
4. The subject does a **standard exercise**, e.g. step-ups at a fixed rate (set by a metronome) for 2 minutes.
5. Reconnect **immediately** and record for 1–2 minutes. Keep recording to find how long breathing takes to return to the resting rate (recovery time).
6. Repeat with other subjects (resting between trials) and calculate means.

Why the values rise in exercise

- Muscles respire faster to make more ATP, so they use **more O_2** and make **more CO_2**.
- CO_2 lowers blood pH. **Chemoreceptors** detect this, and the **ventilation centre** sends more impulses to the diaphragm and intercostal muscles: breathing becomes **deeper** (tidal volume up) and **faster**, so minute ventilation rises.
- O_2 consumption stays above the resting rate for a while after exercise (the **oxygen debt**, 7.5).

Variables, safety and errors

- **Control:** use the same subject, or match age, sex, mass and fitness; same type, intensity and length of exercise; same room temperature.
- **Safety:** sterilise the mouthpiece; use fresh soda lime (otherwise CO_2 builds up and the subject becomes dizzy); stop if the subject feels unwell; soda lime is corrosive.
- **Errors:** leaks at the mouthpiece or nose; breathing unnaturally through a mouthpiece (practise first); exhausted soda lime makes O_2 use look too low; a sloping trace is hard to read by eye (use a digital spirometer, or average several breaths).

MARK-SCHEME WORDING

- Minute ventilation = "tidal volume × breathing rate". Alveolar ventilation uses a formula given in the question.
- Give the s.f. asked: 0.54 breaths s^{-1} × 60 = 32.4 breaths min^{-1}; × 0.04 dm^3 = **1.3 dm^3 min^{-1}**. "1.30" scored only 1 mark (WBI15 Jun 2023).
- O_2 consumption from the **slope** of the trace, with units (e.g. dm^3 s^{-1}).

- Ventilation rises because "chemoreceptors detect the fall in pH / rise in CO_2", "impulses to the intercostal muscles and diaphragm", "muscles need more ATP, so more O_2 for aerobic respiration".

Often asked as: not shown as a trace in any WBI15 paper from 2020 to 2026, so a likely future question; ventilation calculations from given numbers (WBI15 Jan 2022, Jun 2023). Unit 6 asks for trace reading and the plan.

COMMON MISTAKE

- Measuring tidal volume from zero, or from the baseline, to a peak. It's **peak to the next trough**.
- Forgetting why the trace falls: O_2 is used up and the CO_2 is absorbed.
- Not converting cm^3 to dm^3 (÷ 1000), or not rounding to the s.f. asked.

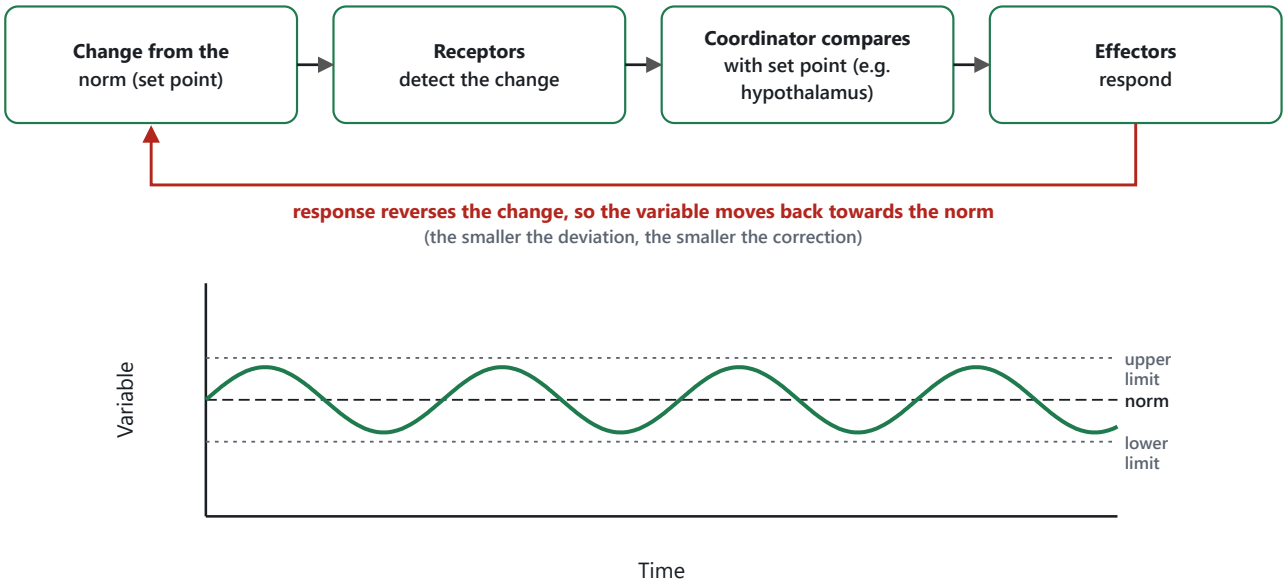
SPEC 7.16 Negative and positive feedback

WHAT THIS SPEC POINT ASKS

(i) Understand what is meant by negative feedback and positive feedback. (ii) Understand how negative feedback keeps systems within narrow limits.

Negative feedback

- A change away from the **norm (set point)** is detected by **receptors**. A **coordinator** (e.g. the hypothalamus) compares it with the set point and sends impulses or hormones to **effectors**. Their response **reverses the change**, bringing the variable back towards the norm.
- Once the variable is back at the norm, the receptors stop detecting a change, so the corrective response is switched off.



Negative feedback. The response reverses the change, so the variable fluctuates a little around the norm but stays between narrow limits.

How negative feedback keeps conditions within narrow limits

- Receptors monitor the variable **all the time**, so even a **small** change is detected and corrected before it grows.
- The bigger the change, the bigger the correction. As the variable nears the norm, the correction shrinks, so there's little overshoot.
- There are usually **separate mechanisms** for correcting a rise and a fall (sweating v shivering; insulin v glucagon). Having both gives more control and a faster response in either direction.
- The result is a **dynamic equilibrium**: the variable fluctuates slightly around the set point but stays within narrow limits.

Positive feedback

- A change triggers a response that **increases the change further**, moving the variable even further from where it started. It **amplifies** the change instead of reversing it.
- It is useful only when a fast, large change is needed and something else then ends it:
  - **Depolarisation of a neurone**: a few Na^+ channels open → Na^+ diffuses in → the membrane depolarises → more voltage-gated Na^+ channels open → more depolarisation, until an action potential is reached.
  - **Childbirth**: contractions of the uterus → more oxytocin released → stronger contractions, until the baby is born.
  - **Blood clotting**: activated platelets release substances that activate more platelets.
- It is harmful when homeostasis breaks down. In hypothermia, a lower body temperature slows metabolism, so less heat is released and the temperature falls further.

WORKED EXAMPLE

Worked example: calculating a range

Blood calcium is kept within 6% of 91 mg dm^{-3}. 6% of 91 = 0.06 × 91 = 5.46.

Range = 91 − 5.46 to 91 + 5.46 = **85.54 to 96.46 mg dm^{-3}** (WBI15 Oct 2024). Calculate **both** limits.

MARK-SCHEME WORDING

- Difference: "positive feedback **increases** the change in the variable; negative feedback **decreases / reverses** the change, returning it to normal".
- Negative feedback (2): "change **detected by a receptor**"; "**effector** stimulated that **counteracts** the change".
- Narrow limits: "constant monitoring", "small deviations detected and corrected", "separate mechanisms to correct a rise and a fall", "response switched off once back at the norm".
- Depolarisation as positive feedback (any 4): "neurotransmitter binds, a few **Na^+ channels open**"; "Na^+ enters and **depolarises** the membrane slightly"; "**more Na^+ channels open**"; "the membrane depolarises further"; "continues until an action potential". Write "sodium ions"; potassium ions are ignored.

Often asked as: describe the difference between negative and positive feedback; explain how negative feedback keeps a system within narrow limits; why depolarisation is positive feedback; 6-mark negative feedback of blood glucose; calculate the limits of a range.

COMMON MISTAKE

- Defining negative feedback, then giving only an **example** for positive feedback with no definition.
- Thinking negative feedback only lowers a variable. It **opposes** a change in **either** direction; positive feedback reinforces it.
- Giving only the detection, or only the effector, in a 2-mark answer.
- Writing "negative feedback" when the question asks for the name of the process (homeostasis, osmoregulation).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Blood glucose control (insulin, glucagon, glycogenesis, glycogenolysis) is the example in a 6-mark negative feedback question. The top level needs the **switch-off**: less insulin (or glucagon) once glucose is back to normal. Don't confuse glucagon with glycogen (WBI15 Jan 2023).
- Positive feedback is examined at the **action potential**, not childbirth or clotting (WBI15 Jan 2026).

SPEC 7.17 Homeostasis and thermoregulation

WHAT THIS SPEC POINT ASKS

Understand what is meant by homeostasis and why it matters for keeping the body in dynamic equilibrium during exercise, including the role of the hypothalamus in thermoregulation.

Homeostasis

- **Homeostasis**: maintaining a **constant internal environment**, within narrow limits, despite changes inside and outside the body. It is a **dynamic equilibrium**: values fluctuate slightly around a set point and are constantly corrected by negative feedback.
- Why it matters:
  - **Temperature**: enzymes work best near 37 °C. Too hot and they **denature**; too cold and reactions slow down.
  - **pH**: changes alter the charges on amino acid R groups, changing the shape of enzymes and other proteins.
  - **Water potential of the blood**: too low and cells **shrink** as water leaves by osmosis; too high and cells swell and may burst.
  - **Blood glucose**: enough for respiration, but not so much that it lowers the water potential of the blood.

Dynamic equilibrium during exercise

Exercise pushes several variables away from the norm at once. Homeostatic mechanisms bring them back:

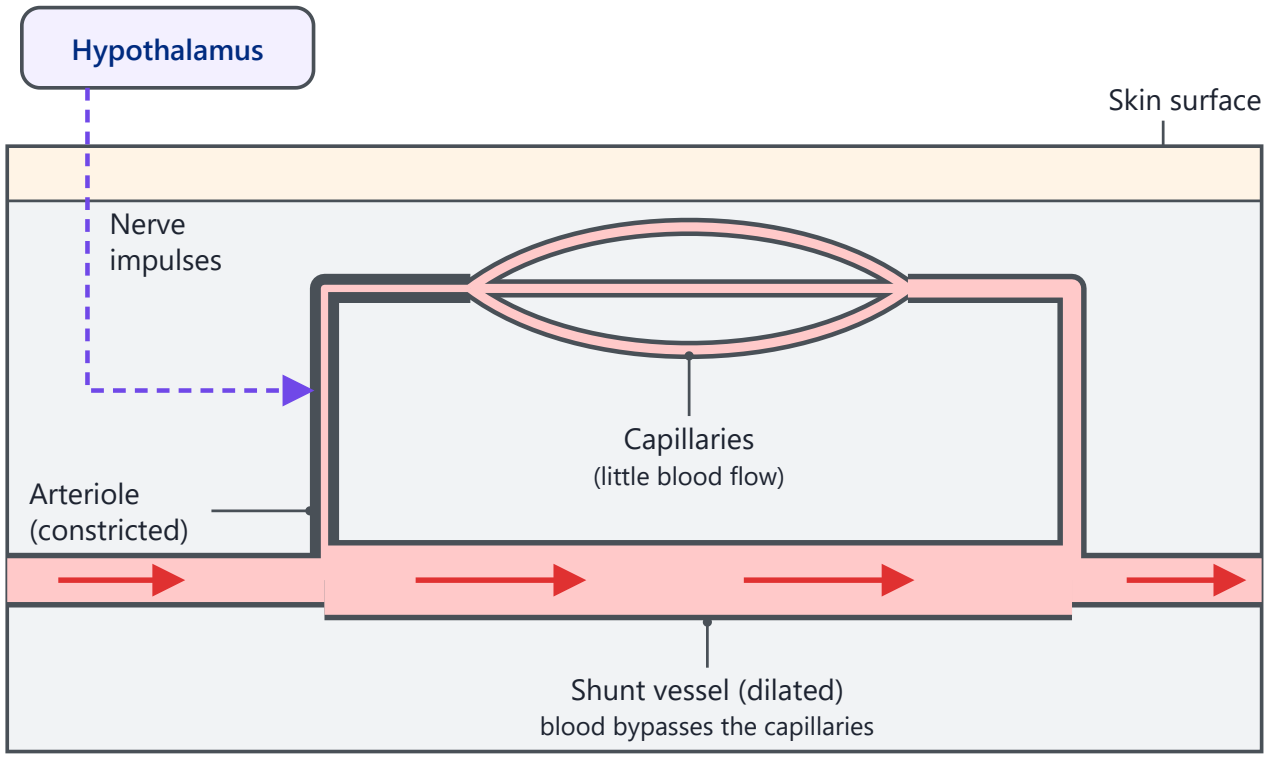
Exercise causes	Detected by	Correction
more heat from respiration: core temperature rises	thermoreceptors → hypothalamus	sweating; vasodilation in the skin
more CO_2: blood pH falls	chemoreceptors → medulla	faster, deeper breathing; higher heart rate
glucose used faster: blood glucose falls	pancreas	glucagon (and adrenaline): liver glycogen broken down to glucose
water lost in sweat: blood water potential falls	osmoreceptors → hypothalamus	more ADH, less urine (7.21)

Thermoregulation: the role of the hypothalamus

- **Thermoreceptors** in the **hypothalamus** detect the temperature of the blood flowing through it (core temperature). Thermoreceptors in the **skin** detect changes in the outside temperature early.

- The **thermoregulatory centre** in the hypothalamus (with a heat loss centre and a heat gain centre) compares the temperature with the set point (about 37 °C) and sends **impulses along motor and sympathetic neurones** to the effectors.

Too hot: heat loss centre	Too cold: heat gain centre
Vasodilation: arterioles supplying the skin capillaries dilate, so more blood flows near the surface and more heat is lost by radiation	**Vasoconstriction**: skin arterioles constrict, so blood is diverted through shunt vessels away from the surface and less heat is lost
Sweating: water evaporates from the skin, using heat energy from the body	Less sweat produced
Hair erector muscles relax: hairs lie flat, trapping less insulating air	Hair erector muscles contract: hairs stand up and trap insulating air (little effect in humans)
Metabolic rate lowered	**Shivering**: rapid, involuntary muscle contractions; respiration releases heat
	More **adrenaline** and **thyroxine** raise the metabolic rate



In cold conditions the arteriole constricts and the blood is diverted through the shunt vessel, which keeps heat in the core.

The second interactive diagram at the foot of the page shows this negative feedback at work.

MARK-SCHEME WORDING

- Homeostasis: "maintaining the body in a state of **dynamic equilibrium**" or "maintaining a **constant internal environment**". A described example is ignored.
- Thermoregulation (any 3–4): "**negative feedback**"; "**thermoreceptors** detect an increase above 37 °C"; "**impulses** to the **thermoregulatory centre / hypothalamus**" ("heat loss centre" accepted); "impulses via **motor / sympathetic neurones** to a **named effector**"; "a response that increases heat loss or reduces heat gain" (vasodilation, sweat glands, hair erector muscles, lower metabolic rate). "Signal" is rejected; "control centre" on its own is ignored.
- Heat loss in exercise (graph): rises, then reaches a maximum and levels off. "Comment" means describe, not explain.

Often asked as: state what homeostasis means; describe the role of the hypothalamus in keeping temperature steady; temperature control during exercise (humans, kangaroo rats); describe heat-loss graphs; how a seizure raises body temperature.

COMMON MISTAKE

- "Capillaries dilate" or "blood vessels move closer to the skin." The **arterioles** dilate or constrict; no vessels move.
- Describing the response (more sweating) without saying **how the effector is stimulated**: impulses along neurones from the hypothalamus.
- "Signals" or "messages" instead of impulses.
- Giving an example of homeostasis instead of the definition, or "keeping conditions constant" with no idea of narrow limits.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The **heat loss centre** of the hypothalamus is credited by name (WBI15 Jan 2024, Oct 2024).
- Blood redistributed by vasodilation to the skin (heat loss) and the muscles (oxygen) during exercise, so less reaches the kidneys (WBI15 Jan 2026).
- Insulin falls during exercise: read the negative values on a "change in insulin" axis carefully, and watch the time units (WBI15 Jan 2025).

SPEC 7.18 Structure of the mammalian kidney

WHAT THIS SPEC POINT ASKS

Know the gross (whole-organ) and microscopic structure of the mammalian kidney.

Gross structure (a kidney cut lengthways)

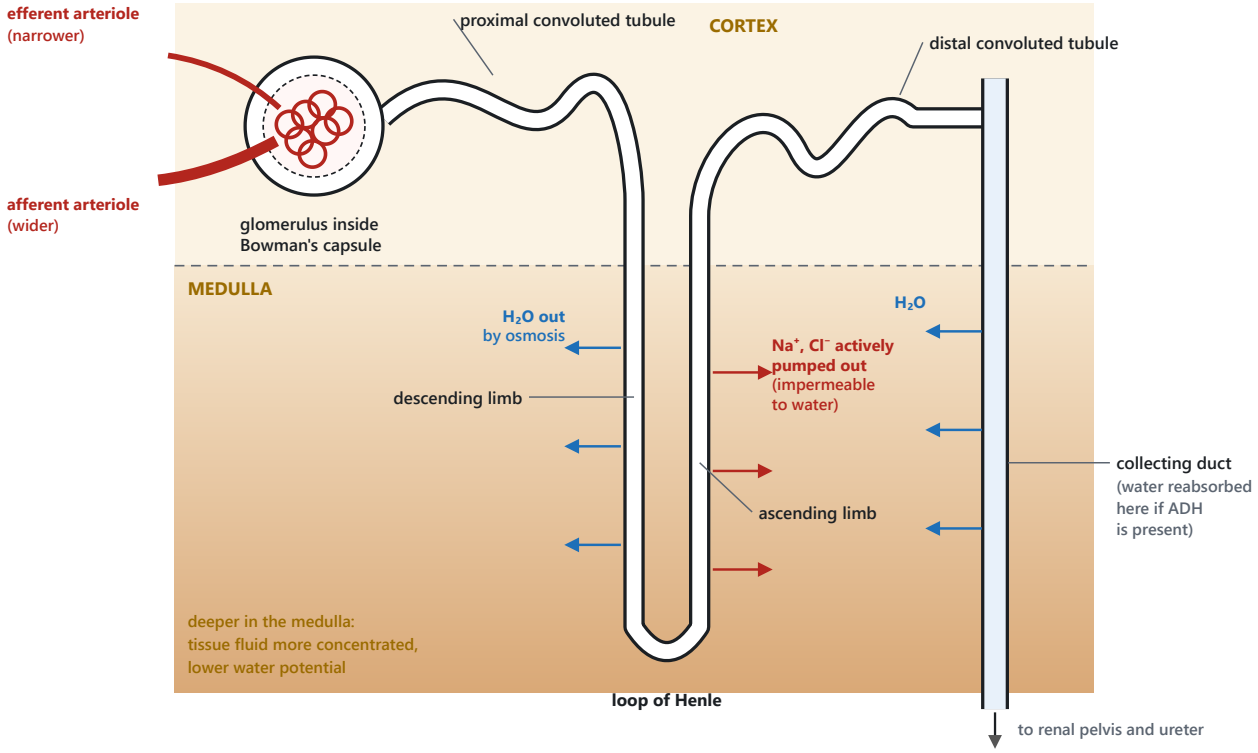
Region	What's there
Fibrous capsule	tough outer layer that protects the kidney
Cortex	outer region: **glomeruli, Bowman's capsules, proximal and distal convoluted tubules**
Medulla	inner region, arranged in cone-shaped pyramids: **loops of Henle** and **collecting ducts** (plus blood vessels)
Renal pelvis	funnel-shaped space where the collecting ducts empty urine; leads into the **ureter**
Renal artery	brings blood (containing urea) from the aorta
Renal vein	takes filtered blood to the vena cava
Ureter	carries urine to the bladder

The nephron

Each kidney contains about a million **nephrons**. Follow the filtrate:

1. **Bowman's capsule** (cortex): a cup around the **glomerulus**, a knot of capillaries. Ultrafiltration happens here (7.19).
2. **Proximal convoluted tubule (PCT)** (cortex): selective reabsorption of glucose, amino acids, most ions and much of the water (7.20).
3. **Loop of Henle** (a descending limb and an ascending limb dipping into the medulla): makes the medulla tissue fluid very concentrated, so water can be reabsorbed later (7.20).
4. **Distal convoluted tubule (DCT)** (cortex): fine-tunes the balance of ions and pH; its water permeability is controlled by ADH.
5. **Collecting duct** (cortex down through the medulla): water is reabsorbed by osmosis when ADH is present. It carries urine to the renal pelvis.

Blood supply: renal artery → **afferent arteriole** (wider) → **glomerulus** → **efferent arteriole** (narrower) → capillaries around the tubules and alongside the loop of Henle → renal vein.



A nephron (schematic, not to scale). Bowman's capsule, the PCT and the DCT are in the cortex; the loop of Henle and collecting duct run down into the medulla. Ions are pumped out of the ascending limb; water leaves the descending limb and the collecting duct by osmosis.

Microscopic features to recognise

Structure	Features	Link to function
Glomerular capillary wall	**endothelium with pores** (fenestrations)	lets plasma through, holds back blood cells
Basement membrane	a mesh of collagen and glycoproteins	the **filter**: stops plasma proteins
Inner wall of Bowman's capsule	**podocytes**: cells with foot-like extensions and **filtration slits** between them	filtrate passes between them into the capsule
PCT cells	**microvilli** (brush border) on the side facing the lumen; many **mitochondria** near the base; infolded basal membrane; tight junctions between cells	large surface area for reabsorption; ATP for active transport

- **Juxtamedullary** nephrons have their glomeruli near the medulla and **long loops** of Henle that reach deep into it. Most nephrons (cortical nephrons) have short loops.

MARK-SCHEME WORDING

- Path of a water molecule: **Bowman's capsule → proximal convoluted tubule → loop of Henle → distal convoluted tubule → collecting duct**.
- Drawing Bowman's capsule: a **cup** around the glomerulus with the PCT leaving it; labels such as efferent arteriole, podocytes, basement membrane, squamous epithelium. Label lines must **end on** the structure. Don't label the structure the question already names.
- PCT cell (3 marks): "**microvilli / brush border** on the **lumen** side give a large surface area"; "**mitochondria** in the **basal** cytoplasm supply ATP"; describing the orientation (apical v basal) is a separate mark.
- "The **basement membrane** prevents plasma proteins from being filtered."

Often asked as: locate the cortex, medulla, pelvis, ureter, renal artery and vein; where Bowman's capsule is (the cortex); draw a labelled Bowman's capsule; which part responds to ADH (collecting duct); PCT cell micrographs; magnification and area calculations.

COMMON MISTAKE

- Leaving the location of Bowman's capsule blank, or placing it in the medulla. It's in the **cortex**.
- Naming microvilli and mitochondria without saying **where** in the cell they are.
- Putting units on a magnification.
- Using the diameter instead of the radius for the area of a vessel (πr^2).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Podocytes, fenestrations, the fibrous capsule and brush-border orientation are credited (WBI15 Jun 2022, Jun 2024).
- Calculations on kidney structures: magnification and volume of a Bowman's capsule (sphere: $\frac{4}{3}\pi r^3$) (WBI15 Oct 2024); area of a renal vein from its diameter (WBI15 Jun 2021).

SPEC 7.19 Urea production and ultrafiltration

WHAT THIS SPEC POINT ASKS

Understand how urea is made in the liver from excess amino acids (no ornithine cycle detail needed) and how it is removed from the blood by ultrafiltration.

Making urea in the liver

- Amino acids can't be stored. Excess amino acids are carried in the blood to the **liver**.
- **Deamination**: the **amine group ($-NH_2$)** is removed, forming **ammonia (NH_3)**. Ammonia is very soluble and very **toxic**.
- The rest of the amino acid (a keto acid) is **respired** (it can enter the Krebs cycle) or converted to **glucose, glycogen or fat** for storage.
- Ammonia is combined with **CO_2** to make **urea**, which is far less toxic, in the **ornithine cycle** (you don't need the details):
  - $2NH_3 + CO_2 \rightarrow CO(NH_2)_2 + H_2O$
- Urea dissolves in the plasma and is carried to the kidneys.

Ultrafiltration in the glomerulus

- The **afferent arteriole** has a **wider** diameter than the **efferent arteriole**. Blood enters the glomerulus faster than it can leave, so a **high hydrostatic pressure** builds up in the glomerular capillaries.
- The pressure forces small molecules out of the blood into the Bowman's capsule, through three layers:
  1. pores (fenestrations) in the capillary **endothelium**,
  2. the **basement membrane** (the actual filter),
  3. the **filtration slits** between the **podocytes**.
- It's filtration **by size, under pressure**. It isn't selective about whether a substance is useful: that's sorted out later by selective reabsorption.

Filtered (in the glomerular filtrate)	Not filtered (stays in the blood)
water, glucose, amino acids, urea, ions (Na^+, Cl^-, K^+), vitamins, creatinine	red and white blood cells, platelets, **large plasma proteins** (albumin, fibrinogen, prothrombin)

- About **125 cm^3** of filtrate is formed per minute (about 180 dm^3 a day), and almost all of it is reabsorbed.

Comparing plasma, filtrate and urine

Substance	Plasma / g dm^{-3}	Glomerular filtrate / g dm^{-3}	Urine / g dm^{-3}
Protein	80	0	0
Glucose	1.0	1.0	0
Urea	0.3	0.3	20
Na^+	3.2	3.2	about 3.5 (varies)

(Typical values.)

WORKED EXAMPLE

Worked example: what the table shows

Protein: in plasma but not in the filtrate, so it is too large to pass through the basement membrane.

Glucose: the same in plasma and filtrate (freely filtered) but absent from urine, so it is **all reabsorbed** (in the PCT).

Urea: urine ÷ filtrate = 20 ÷ 0.3 = **67 times** more concentrated, because most of the **water** has been reabsorbed (even though some urea diffuses back).

Water: 180 dm^3 of filtrate a day but only 1.5 dm^3 of urine: (180 − 1.5) ÷ 180 × 100 = **99.2%** of the water is reabsorbed.

Filtration rate: 6000 cm^3 of filtrate per hour × 24 = 144 000 cm^3 = **144 dm^3** per day (WBI15 Jan 2025).

MARK-SCHEME WORDING

- Excess amino acids (any 3): "transported to the **liver**"; "**amine group removed** by **deamination**"; "converted to **urea**"; "in the **ornithine cycle**". ("Ammonia" and "amino group" are ignored.) The remainder may become keto acids, glucose or pyruvate.
- Ultrafiltration (2–3): "**high (hydrostatic) pressure in the glomerulus**"; "**afferent arteriole** has a **wider** diameter than the **efferent** arteriole"; "small molecules forced through **pores / fenestrations** and the **basement membrane**". "Arteriole" is accepted, "artery" is not. "Pressure" alone isn't enough: say where it is high.
- Large molecules (haemoglobin, prothrombin, plasma proteins) are not filtered; the **basement membrane** is the filter.

Often asked as: describe what happens to excess amino acids; explain how urea (or creatinine) is removed from the blood; the effect of a wider afferent arteriole; which substances are filtered; filtration-rate calculations with unit conversions.

COMMON MISTAKE

- Describing the **digestion** of protein to amino acids when the question asks what happens to the amino acids.
- Vague "pressure" answers that never say the pressure is high **in the glomerulus**, or why (the arteriole diameters).
- Saying urea is "actively secreted" into Bowman's capsule. It is filtered under pressure.
- Unit errors: g → mg is × 1000 (not ÷ 1000); hours → seconds is × 3600.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The ornithine cycle is credited as a mark point even though its detail isn't required (WBI15 Jan 2025).
- **Creatinine**: a small molecule that is filtered and not reabsorbed, so the rate it appears in urine measures the filtration rate (WBI15 Oct 2021).
- Mass of glucose filtered per second, with g → mg and h → s conversions (WBI15 Oct 2024).

SPEC 7.20 Selective reabsorption and the loop of Henle

WHAT THIS SPEC POINT ASKS

Understand how solutes are selectively reabsorbed in the proximal tubule, and how the loop of Henle acts as a countercurrent multiplier to increase the reabsorption of water.

Selective reabsorption in the proximal convoluted tubule

- About **two-thirds** of the water and Na^+ in the filtrate, **all** the glucose and amino acids (normally), and some urea are reabsorbed here.
- How glucose is reabsorbed (**co-transport**, an indirect form of active transport):
  1. **Na^+/K^+ pumps** in the **basal membrane** (the side next to the blood capillary) **actively transport Na^+** out of the PCT cell into the blood, using ATP. This keeps the **Na^+ concentration inside the cell low**.
  2. Na^+ diffuses from the lumen into the cell, down its concentration gradient, through **co-transporter proteins** in the membrane of the **microvilli**. Each carries a **glucose** (or amino acid) molecule with it, against the glucose concentration gradient.
  3. Glucose then leaves the cell on the blood side by **facilitated diffusion**, down its concentration gradient, and enters the capillaries.
- **Cl^-** follows the Na^+. Removing all these solutes lowers the water potential of the tissue fluid and blood, so **water** follows by **osmosis**. Some **urea** diffuses back too.
- Adaptations of PCT cells: **microvilli** (large surface area); **many mitochondria** (ATP for active transport); many co-transporter and carrier proteins; infolded basal membrane; very close to the capillaries.

The loop of Henle: a countercurrent multiplier

- **Ascending limb**: **impermeable to water**. Na^+ and Cl^- are **actively transported out** of it into the tissue fluid of the medulla.
- This **raises the solute concentration** (lowers the water potential) of the tissue fluid in the medulla.
- **Descending limb**: **permeable to water** (it has aquaporins), but much less permeable to ions. Water leaves it by **osmosis** into the concentrated tissue fluid and is carried away in the blood. The filtrate gets more and more concentrated on the way down.
- The filtrate is most concentrated at the hairpin bend. On the way up the ascending limb, ions are pumped out, so it becomes more dilute again.
- **Countercurrent**: the filtrate in the two limbs flows in **opposite directions**, side by side. At any level, the ascending limb can only make a small difference in concentration, but that small difference is **multiplied** along

the length of the loop. The result is a **steep gradient**: the tissue fluid gets more and more concentrated (lower and lower water potential) **deeper** into the medulla.

- Why it matters: the **collecting duct** runs down through this concentrated medulla. Water leaves the collecting duct by **osmosis** all the way down (when ADH has made it permeable), so the urine becomes **concentrated** and water is conserved.
- **Longer loops** → a bigger gradient → more water reabsorbed → **more concentrated urine**. Desert mammals such as the kangaroo rat have very long loops of Henle.

WORKED EXAMPLE

Worked example: glucose in the urine

As plasma glucose rises, the mass of glucose filtered rises in proportion. Reabsorption also rises, until every co-transporter is working at its maximum rate; in WBI15 Jan 2026 this happened when plasma glucose rose above **12 mmol dm^{-3}**.

Above that, glucose is filtered faster than it can be reabsorbed, so the extra stays in the filtrate and appears in the **urine** (as in untreated diabetes).

A rise in plasma glucose from 5 to 35 mmol dm^{-3} is a (35 − 5) ÷ 5 × 100 = **600%** increase (WBI15 Jan 2026).

MARK-SCHEME WORDING

- Glucose reabsorption (any 3): "in the **proximal convoluted tubule**"; "**co-transport** with **sodium ions** using carrier / transport proteins"; "glucose **diffuses** into the blood capillaries down its concentration gradient" (facilitated diffusion accepted). Say "sodium **ions**".
- Countercurrent multiplier (any 4): "**filtrate flows in opposite directions** in the two limbs"; "**ions (Na^+, Cl^-) move out of the ascending limb**"; "by **active transport**"; "**increases** the ion concentration / **decreases** the water potential of the surrounding fluid"; "**water leaves the descending limb** by **osmosis** (through aquaporins)". A concentration gradient "formed" along the loop is accepted.
- Desert mammals: "**longer loop of Henle** (more juxtamedullary nephrons)", "more Na^+ pumped out of the ascending limb", "higher solute concentration in the medulla", "more water reabsorbed", "more mitochondria in the ascending limb". Behaviour (nocturnal, burrowing) is credited too.

Often asked as: describe how glucose is reabsorbed; explain how the loop of Henle acts as a countercurrent multiplier; adaptations of kangaroo rats, camels or other desert mammals; plasma–filtrate–urine data; graphs of glucose filtration v reabsorption.

COMMON MISTAKE

- Explaining water movement well but being vague about the multiplier. Say which limb is permeable to what, and which way the ions move.
- "Sodium" instead of "sodium **ions**". "The tubes flow in opposite directions": it's the **filtrate** that flows.
- "There is a concentration gradient." Say the loop **forms or increases** one.
- Reversing the link: a high solute concentration means a **low** (more negative) water potential.

- Listing differences between two species without a reason for each.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Secretion** of substances into the tubule is credited in a data question on urine composition (WBI15 Jun 2021).
- Desert v wet-habitat mammals: medulla thickness, urine concentration, metabolic water, percentage calculations (WBI15 Oct 2020, Oct 2022, Jan 2024, Oct 2025).

SPEC 7.21 ADH: controlling plasma concentration and blood volume

WHAT THIS SPEC POINT ASKS

Understand how osmoreceptors in the hypothalamus, the pituitary gland and antidiuretic hormone (ADH) control the solute concentration of the plasma and the blood volume by negative feedback.

When the blood's water potential falls (sweating, not drinking, salty food)

1. **Osmoreceptors** in the **hypothalamus** detect the fall in water potential: water leaves them by osmosis and they **shrink**.
2. They stimulate **neurosecretory cells** in the hypothalamus. ADH is **made** in these cells, travels down their axons, and is **released** from the **posterior pituitary gland** into the blood. More ADH is released.
3. ADH travels in the blood to the **distal convoluted tubule** and **collecting duct**, and binds to **receptors** on the cell surface membranes of their cells.
4. This activates a **second messenger** (cAMP), which makes vesicles containing **aquaporins** (water channel proteins) **fuse with the membrane** facing the lumen.
5. The walls become **more permeable to water**. More water leaves the filtrate by **osmosis** into the concentrated medulla, and back into the blood.
6. A **small volume of concentrated urine** is produced. The blood's water potential rises back to normal, the osmoreceptors are no longer stimulated, and less ADH is released: **negative feedback**.

- Thirst is stimulated at the same time, so you drink more.

When the blood's water potential rises (drinking a lot of water)

- The osmoreceptors swell, and **less ADH** is released. Aquaporins are taken back into the cytoplasm in vesicles, so the DCT and collecting duct become **less permeable** to water. Less water is reabsorbed, giving a **large volume of dilute urine**, and the water potential falls back to normal.

Blood volume

- Reabsorbing more water also raises the **blood volume** (and blood pressure). If blood volume falls (after bleeding or heavy sweating), **baroreceptors** in the aortic arch and carotid arteries detect the lower pressure, and more ADH is released. A rise in blood volume reduces ADH release.

MARK-SCHEME WORDING

- Control chain (any 4): "**osmoreceptors** in the **hypothalamus** detect the **water potential** of the blood"; "impulses to the **pituitary gland**, which releases **more / less ADH**"; "ADH acts on the **distal convoluted tubule / collecting duct**"; "**aquaporins** inserted into the membrane / **increased permeability** to water"; "**more / less water reabsorbed**, restoring the blood water potential to normal".
- Too much water: "less ADH", "fewer aquaporins", "DCT / collecting duct less permeable", "more dilute urine".
- High blood volume is detected by **baroreceptors** in the carotid artery or aortic arch.
- Graphs: as ADH rises, urine concentration rises and the rate of urine production falls.
- "Concentration" on its own is ignored: use "**water potential**". "Tubules" on its own is ignored: name the **DCT or collecting duct**.

Often asked as: where ADH is produced and where it acts; name the receptor (osmoreceptor); describe how ADH changes the blood's water potential; the response to drinking a lot of water; keeping blood sodium within narrow limits; urine volume and concentration data after drinking water or salt solution; MDMA and ADH (6-mark).

COMMON MISTAKE

- Writing **chemoreceptors** instead of **osmoreceptors**.
- "ADH makes the collecting duct less permeable" or "more permeable means more urine". It's the reverse.
- Leaving out the pituitary gland, or the ending: the blood's water potential **returns to normal**.
- "ADH affects ultrafiltration." It affects water **reabsorption**.
- Explaining the aquaporin mechanism in detail for a 2-mark "describe" question.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Hyponatraemia**: too much ADH means too much water is reabsorbed, so the blood Na^+ concentration falls and water enters brain cells by osmosis (WBI15 Jun 2024).
- **MDMA** (ecstasy) increases ADH release, so less urine is made and water is retained (WBI15 Oct 2020, 6-mark).
- Why plasma solute concentration must stay within narrow limits: cells would shrink, or swell and burst, if the plasma's water potential changed (WBI15 Jan 2025).
- Aldosterone is accepted as another hormone controlling blood sodium (WBI15 Jan 2025).

SPEC 7.22 Switching genes on and off: transcription factors and hormones

WHAT THIS SPEC POINT ASKS

Understand how genes are switched on and off by transcription factors, including the role of peptide hormones acting from outside the cell and steroid hormones acting inside it.

Transcription factors

- A **transcription factor** is a **protein** that binds to a specific DNA base sequence, usually in the **promoter** region just before a gene, and controls whether that gene is transcribed.
- **Activators** help **RNA polymerase** bind to the promoter. The gene is transcribed into mRNA, which is translated into a protein: the gene is **switched on**.
- **Repressors** block the promoter (or stop activators binding). RNA polymerase can't bind, so the gene isn't transcribed: it is **switched off**.
- Every cell has the same genes. Which transcription factors are active decides which genes are expressed. That's how cells **differentiate**, and how they respond to signals such as hormones.

Steroid hormones: act inside the cell (e.g. oestrogen, testosterone)

1. Steroids are **lipid-soluble**, so they **diffuse through the phospholipid bilayer** into the cell.
2. In the cytoplasm or nucleus, the hormone binds to a specific **receptor protein**, forming a **hormone–receptor complex**.
3. The complex acts as a **transcription factor**: in the nucleus, it binds to the promoter of its target genes.
4. It **activates** (helps RNA polymerase bind) or **represses** transcription of those genes, so more (or less) of particular proteins is made.

- Example: oestrogen binds to its receptor; the complex switches on genes that make cells of the uterus lining divide.

Peptide hormones: act from outside the cell (e.g. insulin, glucagon, ADH)

1. Peptide hormones are **water-soluble** and too large and charged to cross the membrane.
2. The hormone binds to a specific **receptor on the cell surface membrane** (with a complementary shape).
3. This activates a **second messenger** (such as **cAMP**) inside the cell, which starts a **cascade** of enzyme activations that amplifies the signal.
4. The cascade can **activate (or inactivate) transcription factors** in the cytoplasm. Activated transcription factors move into the nucleus and switch their target genes on or off.

- Not every effect of a peptide hormone involves genes. ADH inserting aquaporins (7.21) happens without any new transcription.

	Steroid hormones	Peptide hormones
Examples	oestrogen, testosterone, progesterone	insulin, glucagon, ADH (adrenaline, an amine, acts the same way)
Soluble in	lipids	water
Cross the membrane?	**yes**	**no**
Receptor	**inside** the cell (cytoplasm or nucleus)	on the **cell surface membrane**
Effect on genes	the hormone–receptor complex **is** the transcription factor	a second messenger cascade **activates** a transcription factor
Speed	slower to start, long-lasting	faster

MARK-SCHEME WORDING

- Switching a gene **off** (any 3): "transcription factors are **proteins**"; "**bind to the promoter** / gene"; "cause methylation or deacetylation of histones" (ACCEPT "DNA wrapped more tightly"); "**preventing RNA polymerase binding**".
- Steroid hormones (any 3): "**diffuse across the membrane** into the cell / nucleus"; "bind to an **intracellular receptor**"; "the **hormone–receptor complex acts as a transcription factor**"; "activates or represses transcription". "Binds to cell membrane receptors" is rejected. Don't just repeat "regulate transcription" from the stem.
- Peptide hormone, growth factor or cytokine (any 3): "binds to a **receptor on the cell surface membrane**"; "activates a **second messenger** / signalling cascade and a **transcription factor**"; "the transcription factor **binds to the promoter**, so **RNA polymerase** binds"; "the gene is transcribed and the mRNA translated into the protein".
- Name the protein made when the context gives one (actin, myosin, troponin or tropomyosin for muscle; the enzymes that make dopamine).
- Use "fewer / reduced" when the question gives a fall: less testosterone enters cells → **fewer** hormone–receptor complexes → **less** transcription.

Often asked as: describe how genes are switched off by transcription factors; describe how steroid hormones regulate transcription; hormones, growth factors or cytokines leading to differentiation (muscle cells, stem cells, T cells); oestrogen and cell division; testosterone and muscle protein.

COMMON MISTAKE

- Describing genes being switched on and off without ever mentioning **transcription factors**.
- Saying a peptide hormone (insulin, glucagon) enters the nucleus or **is** a transcription factor. Peptide hormones act at the cell surface (one mark scheme slipped on this: don't copy it).

- "Transcription factor → mRNA → dopamine." The mRNA is translated into an **enzyme**, and the enzyme makes dopamine.
- Generic answers that ignore the cell type or hormone named in the question.
- Copying phrases from the stem or the article.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Epigenetics**: methylation of DNA (at cytosine bases) and deacetylation of histones pack DNA tightly and silence genes. **Histone acetylation** removes positive charge, so DNA is wrapped less tightly and transcription can happen (WBI15 Jan 2023, Jan 2024).
- Genomic imprinting and Prader–Willi syndrome (WBI15 Jan 2024).
- Alternative splicing producing several forms of a receptor from a few genes (WBI15 Jun 2023).

Quick checks

QUICK CHECK

What is the main purpose of the Bainbridge reflex?

- To lower heart rate after exercise
- To raise heart rate when venous return increases, limiting build-up of venous pressure
- To constrict the arterioles in the skin
- To stop breathing during exercise

Increased venous return stretches the atrial walls, which triggers faster heart rate through the sympathetic system. The first option describes the baroreceptor response instead, which is the commonest confusion between the two reflexes.

QUICK CHECK

Which change would NOT help a person who has become too cold?

- Hairs stand up to trap air
- Arterioles in the skin constrict
- Shivering increases
- Sweat glands are stimulated to cool the skin

In cold conditions sweating is inhibited, because evaporation would lose heat. The other three responses all conserve or produce heat.

QUICK CHECK

What is the effect of a high blood CO_2 level on breathing?

- Breathing rate falls to save oxygen
- Chemoreceptors signal the inspiratory centre, so breathing rate and depth increase
- Stretch receptors stop the expiratory centre
- The diaphragm relaxes completely

Raised CO_2 lowers blood pH, which the chemoreceptors detect. They send more impulses to the inspiratory centre, so breathing speeds up and deepens. Stretch receptors work on lung inflation, not on CO_2 levels.

QUICK CHECK

Which example is a positive feedback mechanism?

- Control of body temperature by sweating
- Baroreceptor control of blood pressure
- Platelets releasing a chemical that activates more platelets during clotting
- Breathing rate controlled by stretch receptors

In clotting, each change reinforces the next, which is what positive feedback does. The other three options all bring a variable back towards a set point, which makes them negative feedback systems.